

Matthew Beirouti

TECHNICAL SKILLS:

Coding: C#, C/C++, Java, Python, Powershell, Go, Ruby (Rails), HTML, CSS, JavaScript, React, GraphQL, KQL / SQL, VHDL
Tools / OSes: Git, IntelliJ, Visual Studio / VS Code, .NET, Spring, Docker, Kubernetes, Travis, LaunchDarkly, Azure, AWS, Scikit, OpenCV, LaTeX, Microsoft Office, macOS, Windows, Linux

EDUCATION:

McGill University, Montreal, QC

Bachelor of Engineering - Computer Engineering

May 2018

Noteworthy Classes: *Applied Machine Learning (COMP 551, grad level)*, *Artificial Intelligence (COMP 424)*, *Intro to Computer Vision (ECSE 415)*, *Leadership (ORGB 321)*, *Strategies for Sustainability (MGPO 440)*

WORK EXPERIENCE:

Microsoft, Redmond, WA

November 2023 – Present

Software Engineer (C# / C++) – Cloud Hardware Provisioning

- Led migration of critical data deployments from a high-risk “blastout” model to a progressive, gated rollout system with data-driven auto-gating, reducing failure impact radius and enabling safe, controlled releases across millions of machines
- Built and drove adoption of a self-serve CLI used by ~10 engineers that automates recurring workflows, reducing manual effort by >90%, minimizing human error and improving developer experience; authored docs to streamline onboarding
- Introduced structured logging, metrics, and experimentation instrumentation to improve observability of provisioning workflows, enabling data-driven insights and faster detection of issues in production
- Designed and implemented Provisioning Service, a core distributed component of the hardware provisioning workflow used across millions of machines, enforcing certification-based access controls and eliminating key cloud security risks
- Drove cross-team design and feature decisions for Provisioning Service by collecting and analyzing data, authoring design proposals, and aligning with networking, deployment, and other partner teams to influence architecture and rollout strategy
- Designed and implemented a secure C++ client for Provisioning Service, ensuring safe and reliable integration within the provisioning pipeline
- Enhanced an integration testing framework with automation and usability improvements (config standardization, script automation, new features), reducing test setup time by 25% and accelerating PR validation cycles

Amazon, Seattle, WA

March 2020 – October 2023

Software Development Engineer (Java) – Customer Data Foundations / AmazonAPI

- Partnered with PMs to instrument a Tier-1 service with metrics that surfaced customer usage patterns, enabling product and platform decisions based on real behavioral data
- Designed and implemented an API enabling services on an internal serverless compute platform to integrate with our Tier-1 service, unblocking VP-level goals
- Collaborated to redesign and refactor the backend services powering the delivery block feature on the retail website, reducing operational risk and development complexity for projected savings of \$500K+ and ~1 dev-year annually
- Co-designed and implemented an automated onboarding experience (models, views, and controllers) for a newly developed API, enabling self-serve onboarding at scale
- Led biweekly operational excellence reviews to identify service anomalies and gaps, driving improvements to metrics, alarms, dashboards, and runbooks for our Tier-1 service
- Organized and led performance and stress testing for our largest yearly peak, reducing testing operating costs by ~40%
- Independently migrated our Tier-1 service codebase and fleet from JDK 8 to JDK 11 and from x86 to aarch64

Cogo Labs, Cambridge, MA

October 2018 – November 2019

Software Engineer (Golang) – Platform Team

- Collaborated to architect and develop a high throughput microservice API which enables analysts to interface with hundreds of databases, 5x faster per-request performance than the previous API
- Independently built the full stack of an in-house email validation tool, replacing a third-party vendor and reducing costs by over \$500K per year
- Added accessibility features informed by collected user feedback, improving usability for the analyst user base